VARDHAMAN
COLLEGE OF ENGINEERING

A Course End Project Report on

Waste Segregation and Recycling Analysis System

A8514- Database Management System Laboratory

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Abstract

The **Waste Segregation and Recycling Analysis System** is a database-driven solution designed to streamline the management of solid waste. It focuses on collecting, storing, and analyzing data related to different types of waste, including biodegradable, non-biodegradable, and recyclable materials. By integrating a robust database management system (DBMS), the system ensures that all information regarding waste quantity, source, and category is efficiently recorded and easily retrievable for processing and reporting.

The system leverages automated data capture techniques, such as sensors and image recognition, to accurately identify and categorize waste. These real-time inputs are fed into the database, where structured tables and relationships enable quick analysis, tracking, and reporting. Advanced queries allow administrators to generate insights into waste generation patterns, identify high-volume areas, and monitor recycling efficiency. This DBMS-centric approach reduces human error, improves data consistency, and supports informed decision-making for waste management authorities.

Furthermore, the system provides a platform for generating reports and analytics that promote sustainable practices. By analyzing historical data and trends, organizations can optimize collection schedules, design effective recycling programs, and reduce environmental impact. Ultimately, the **Waste Segregation and Recycling Analysis System** demonstrates how database management technologies can contribute to environmental sustainability by enhancing operational efficiency, minimizing landfill waste, and supporting a cleaner community.

Keywords:

Waste Management, Waste Segregation, Recycling, Smart Campus, Database System, Data Analysis, Sustainability, Environmental Monitoring

Introduction

Waste management is an important aspect of maintaining a clean and healthy environment, especially in campuses and urban areas. With the increasing population and daily activities, large amounts of waste are generated in different forms such as wet waste, dry waste, and plastic waste. Proper segregation and tracking of this waste is necessary to ensure effective recycling and disposal. Traditional waste management methods often lack proper monitoring and data recording, which leads to inefficient handling and increased environmental impact. This project focuses on developing a Smart Waste Tracking System using database management concepts to efficiently store and manage waste-related data.

The system is based on fundamental concepts of Database Management Systems (DBMS) such as data organization, relational tables, and structured queries. It uses tables to store information about waste types, collection locations, quantities, and recycling status. Basic SQL operations like INSERT, SELECT, UPDATE, and DELETE are used to manage the data, while aggregate functions such as SUM and COUNT help in analyzing waste generation and recycling patterns. By applying these concepts, the system provides useful insights like identifying high waste-generating areas and calculating recycling efficiency. This project demonstrates how database systems can be used for real-world applications and supports better decision-making for sustainable waste management.

Literature Review

Waste management and recycling have been widely studied in recent years due to increasing environmental concerns and rapid urbanization. Many research works and reports highlight the importance of proper waste segregation at the source to improve recycling efficiency and reduce landfill usage. Traditional waste management systems mainly focus on manual collection and disposal, which often lack proper tracking and data analysis. Studies suggest that maintaining systematic records of waste generation can significantly improve planning and resource utilization.

Existing methodologies for waste management include basic record-keeping systems and some semi-automated solutions used in municipalities and institutions. These systems typically store data related to waste type, quantity, and collection schedules. However, many of them do not provide detailed analytical insights such as identifying high waste-generating areas or measuring recycling performance effectively. Some advanced solutions integrate technologies like IoT for real-time monitoring, but they are complex and costly to implement for small-scale environments like campuses.

The proposed Smart Waste Tracking System focuses on a simpler and more practical database-driven approach. By using fundamental DBMS concepts, the system efficiently stores and manages waste data and provides useful analysis through structured queries. This approach bridges the gap between basic manual systems and complex automated solutions, making it suitable for educational institutions. It helps in understanding waste patterns, improving recycling practices, and supporting sustainable environmental management.

Objectives

- To design and develop a database system for recording and managing different types of waste generated in various locations.
- To analyze waste data using SQL queries in order to identify waste patterns and measure recycling efficiency.
- To provide a simple and effective solution for improving waste management and promoting environmental sustainability.

1 Methodology

1.1 Architectural Diagram

DESCRIPTION:

This diagram represents the database schema for a waste segregation and recycling analysis system. It consists of four main tables: locations, waste_collection, waste_types, and recycling. The locations table stores unique location identifiers and their names where waste is collected. The waste_types table defines different categories of waste. The waste_collection table records each collection event, linking the location, type of waste, quantity collected, and the date, serving as the central hub connecting locations and waste types. Finally, the recycling table tracks the recycling process, specifying the quantity of collected waste that is actually recycled, referencing the waste_collection entries. Together, this structure enables comprehensive tracking and analysis of waste generation, segregation, and recycling efficiency across multiple locations and waste categories.

Hardware and Software Requirements

Hardware Requirements:

- Computer/Laptop with minimum 4 GB RAM
- Processor: Intel i3 or above
- Hard Disk: 500 GB or above
- Keyboard and Mouse

Software Requirements:

- Operating System: Windows 10/11 or Linux
- Database: MySQL / SQLite
- Frontend (optional): HTML, CSS
- Tool: MySQL Workbench / XAMPP / phpMyAdmin

Working Principle

1. The Smart Waste Tracking System works by collecting and storing data related to waste generated in different locations such as hostels or campus blocks. The user (admin) enters details like waste type (wet, dry, plastic), quantity, and date into the system. This data is stored in a structured database using relational tables. Each table is connected using keys, which helps in organizing and retrieving data efficiently.

2. The system uses basic Database Management System (DBMS) concepts such as data storage, relationships, and query processing. SQL queries are used to perform operations like inserting new data, updating existing records, and retrieving required information. Aggregate functions such as SUM and COUNT are used to calculate total waste, identify the most generated waste type, and measure recycling efficiency.

3. Based on the stored data, the system generates useful outputs such as reports on waste generation, recycling status, and location-wise analysis. These results help in understanding waste patterns and improving waste management practices. Thus, the system provides a simple and effective way to track waste and support better decision

Software Implementation

Program (SQL Code)

```
CREATE DATABASE waste_management;
USE waste_management;

CREATE TABLE locations (
 location_id INT PRIMARY KEY,
    location_name VARCHAR(50)
);

CREATE TABLE waste_types (
 waste_type_id INT PRIMARY KEY,
    waste_type VARCHAR(50)
);

CREATE TABLE waste_collection (
    collection_id INT PRIMARY KEY,
    location_id INT,
    waste_type_id INT,
    quantity INT,
    date DATE,
    FOREIGN KEY (location_id) REFERENCES locations(location_id),
    FOREIGN KEY (waste_type_id) REFERENCES waste_types(waste_type_id)
);

CREATE TABLE recycling (
    recycle_id INT PRIMARY KEY,
    collection_id INT,
    recycled_quantity INT,
    FOREIGN KEY (collection_id) REFERENCES waste_collection(collection_id)
);

INSERT INTO locations VALUES
(1, 'Hostel A'),
(2, 'Hostel B'),
(3, 'Canteen');

INSERT INTO waste_types VALUES
(1, 'Wet Waste'),
(2, 'Dry Waste'),
(3, 'Plastic');

INSERT INTO waste_collection VALUES(1, 1, 1, 10, '2026-04-01'),
(2, 2, 2, 5, '2026-04-01'),
(3, 3, 3, 8, '2026-04-02'),
(4, 1, 2, 6, '2026-04-02');
```

```
INSERT INTO recycling VALUES
(1, 1, 6),
(2, 2, 3),
(3, 3, 5),
(4, 4, 4);

SELECT SUM(quantity) AS total_waste
FROM waste_collection;

SELECT l.location_name, SUM(wc.quantity) AS total
FROM waste_collection wc
JOIN locations l ON wc.location_id = l.location_id
GROUP BY l.location_name;

SELECT wt.waste_type, SUM(wc.quantity) AS total
FROM waste_collection wc
JOIN waste_types wt ON wc.waste_type_id = wt.waste_type_id
GROUP BY wt.waste_type
ORDER BY total DESC
LIMIT 1;

SELECT
(SUM(r.recycled_quantity) / SUM(wc.quantity)) * 100 AS recycling_percentage
FROM waste_collection wc
JOIN recycling r ON wc.collection_id = r.collection_id;
```

Description

The above program implements the Smart Waste Tracking System using SQL. It begins by creating a database and defining tables to store information about locations, waste types, waste collection, and recycling details. Sample data is inserted into each table to represent real-world waste records.

The system uses SQL queries to perform analysis on the stored data. It calculates the total waste generated, identifies the waste generated by each location, and determines the most common type of waste. It also calculates recycling efficiency by comparing recycled waste with total waste. These operations help in organizing the data and providing meaningful insights for better waste management.

2 Results and Discussion

Total Waste Generated

total_waste
29

Waste Generated by Each Location

location_name	total
Hostel A	16
Hostel B	5
Canteen	8

Most Generated Waste Type

waste_type	total
Wet Waste	10

Total Wastage (Not Recycled)

total_wastage
11

3 Mapping POs and SDGs

3.1 Program Outcomes (POs) Mapping

Table 1: Mapping of Program Outcomes (POs) to Project Relevance

PO No.	Program Outcome	Relevance
PO1	Engineering Knowledge	Applies core engineering and DBMS concepts to manage waste data.
PO2	Problem Analysis	Identifies and analyzes issues in waste segregation and recycling.
PO3	Design/Development of Solutions	Develops a system for automated waste categorization and tracking.
PO4	Conduct Investigations of Complex Problems	Investigate waste patterns using data collection and analytics.
PO5	Engineering Tool Usage	Uses DBMS, programming, and sensor tools for implementation.

3.2 Sustainable Development Goals (SDGs) Mapping

Table 2: Mapping of Sustainable Development Goals (SDGs) to Project Relevance

SDG No.	Goal	Relevance
SDG 3	Good Health and Well-being	Reduces health hazards from improper waste disposal and promotes cleaner environments.
SDG 6	Clean Water and Sanitation	Prevents water pollution caused by unmanaged waste and improves sanitation.
SDG 7	Affordable and Clean Energy	Supports efficient energy use by promoting recycling and resource recovery.
SDG 8	Decent Work and Economic Growth	Creates opportunities in waste management, recycling, and green technologies
SDG 9	Industry, Innovation and Infrastructure	Encourages innovative solutions like sensor-based segregation and data-driven waste management.
SDG 11	Sustainable Cities and Communities	Promotes sustainable urban waste practices for cleaner, safer cities.
SDG 12	Responsible Consumption and Production	Encourages recycling, resource optimization, and reduction of landfill waste.
SDG 13	Climate Action	Reduces environmental impact and greenhouse gas emissions from waste.

Conclusion

The **Waste Segregation and Recycling Analysis System** successfully demonstrates how database management, automation, and analytics can improve solid waste management. By integrating sensors, image recognition, and a structured DBMS, the system efficiently categorizes waste, records data, and provides actionable insights for recycling and disposal. Its implementation reduces human effort, minimizes errors in segregation, and supports informed decision-making for authorities and organizations. Overall, the project contributes to environmental sustainability, cleaner communities, and efficient resource utilization, showing the practical impact of combining technology with waste management practices.

Future Work / Recommendations:

Future enhancements could include integrating **IoT-based real-time monitoring** for waste bins across multiple locations, developing a **mobile app interface** for citizens and authorities, and implementing **machine learning algorithms** to improve waste recognition accuracy. Additionally, incorporating predictive analytics can help forecast waste generation patterns, optimize collection schedules, and enhance recycling efficiency. Expanding the system to manage industrial and hazardous waste and linking it with smart city infrastructure would further increase its scalability and environmental impact.

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